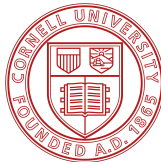


Mapping the Tesseral Field of Saturn

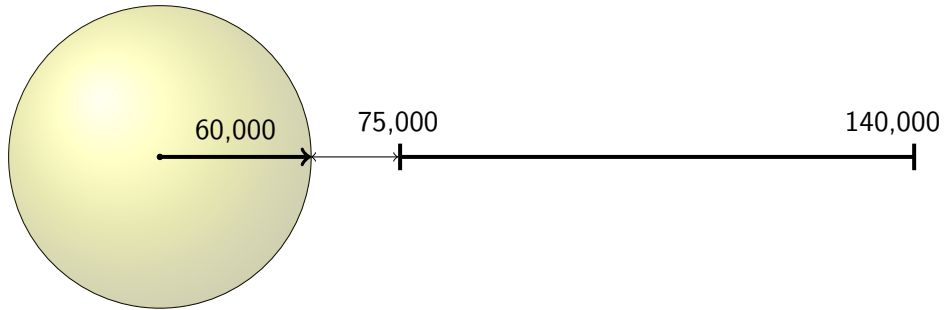
An analysis of Saturnian resonances

Aurora Hiveley
Adv. Maryame El Moutamid

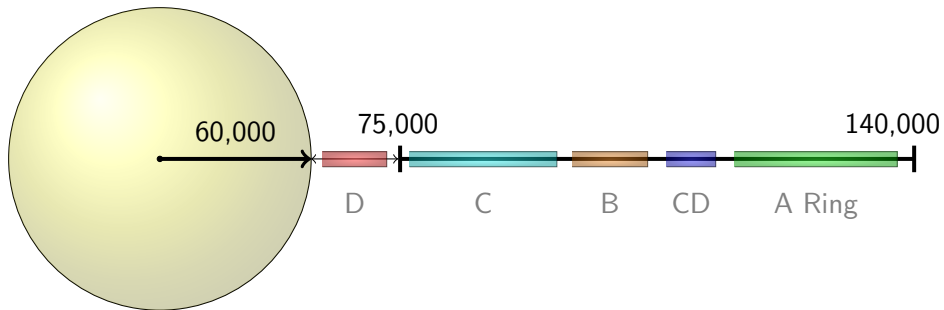
Cornell Center for Astrophysics and Planetary Science
Cornell University

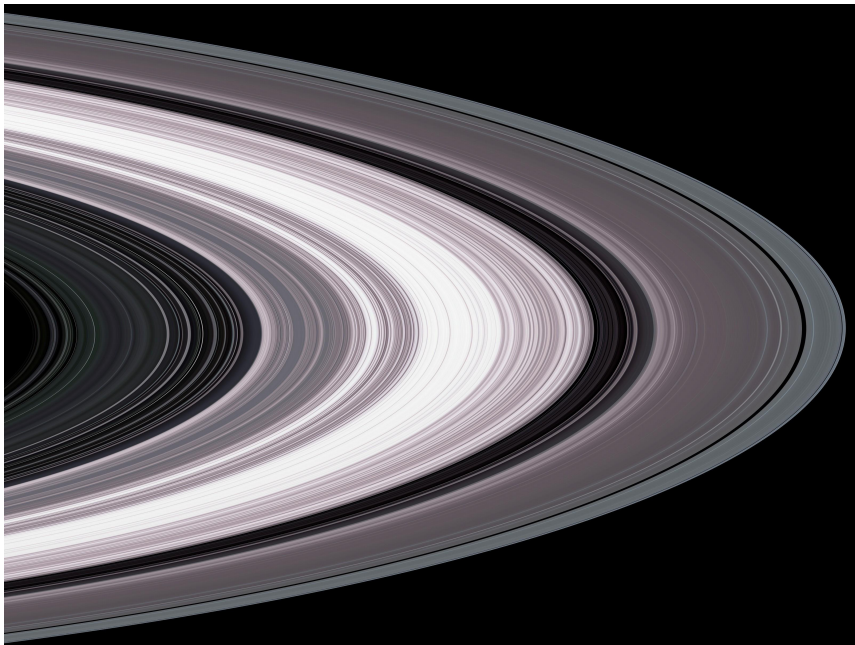


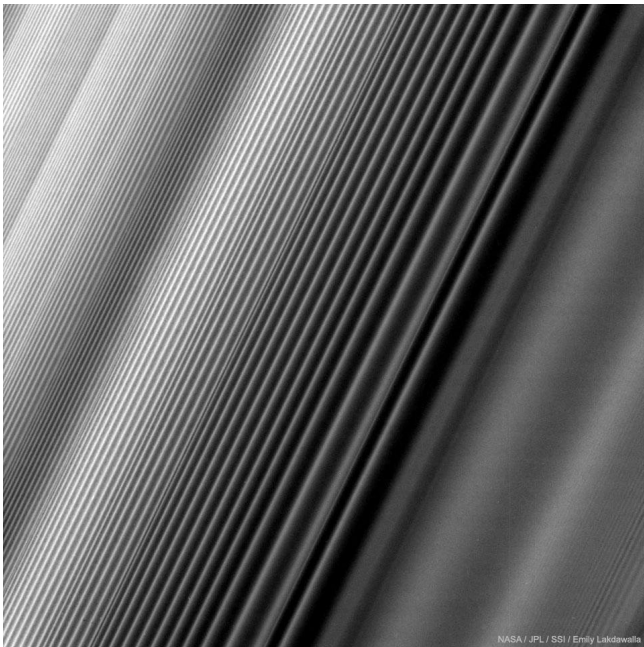
Saturnian Rings



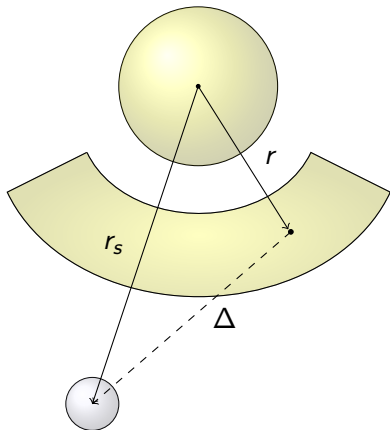
Saturnian Rings



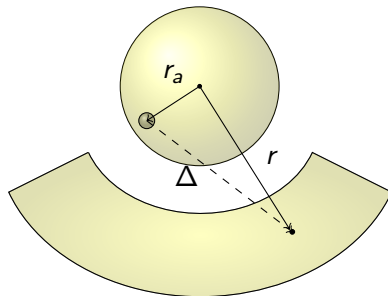
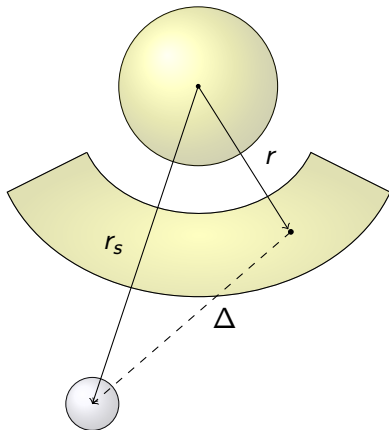




Density Waves

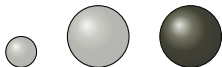


Density Waves

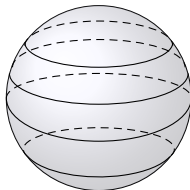


Anomaly Properties

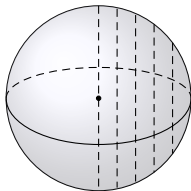
Mass



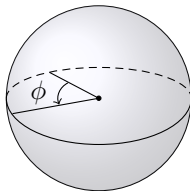
Latitude



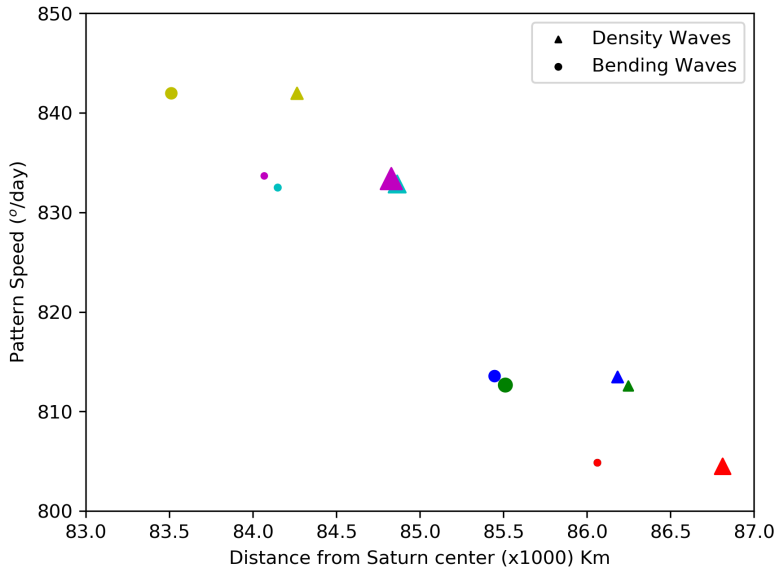
Depth



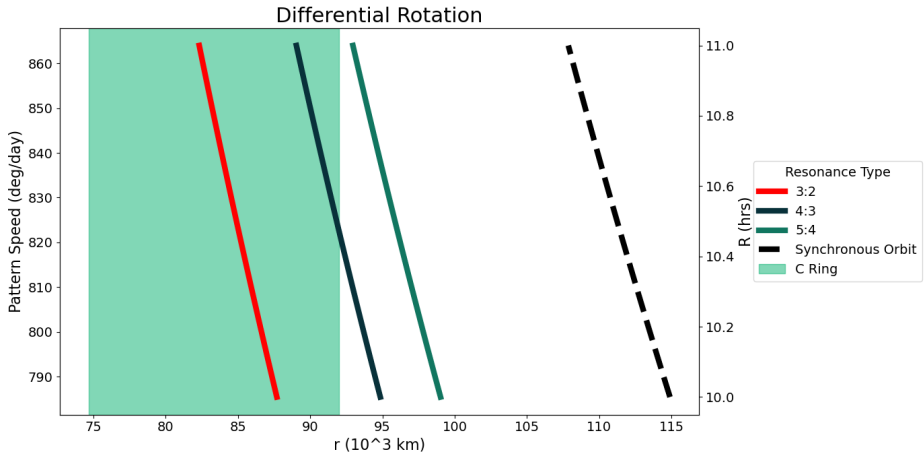
Phase



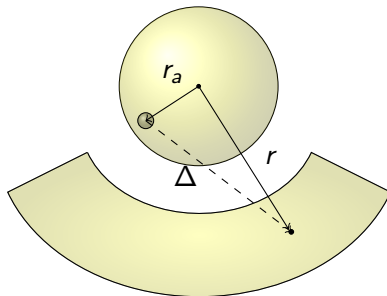
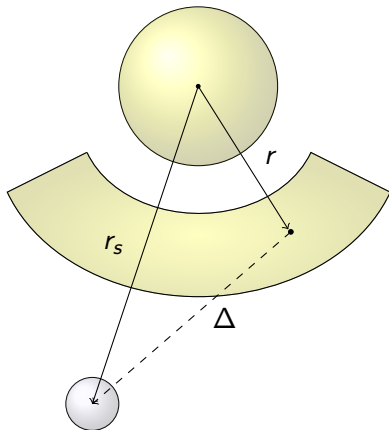
Data Analysis



Resonances

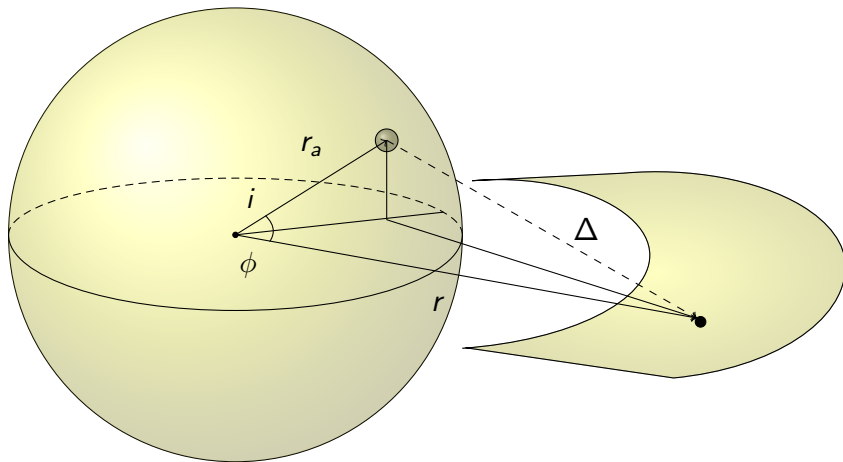


Density Waves

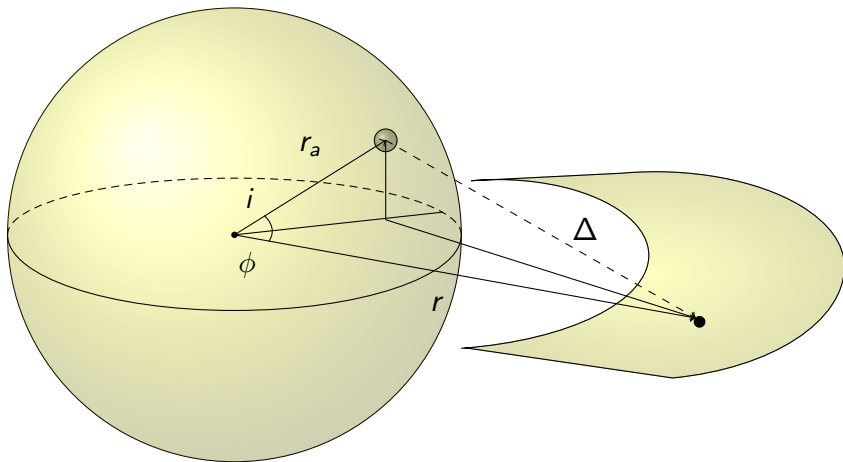


$$F_G + \text{Resonances} \Rightarrow f(\text{Radius, Mass}) = \text{Amplitude}_{\text{DW}}$$

Bending Waves

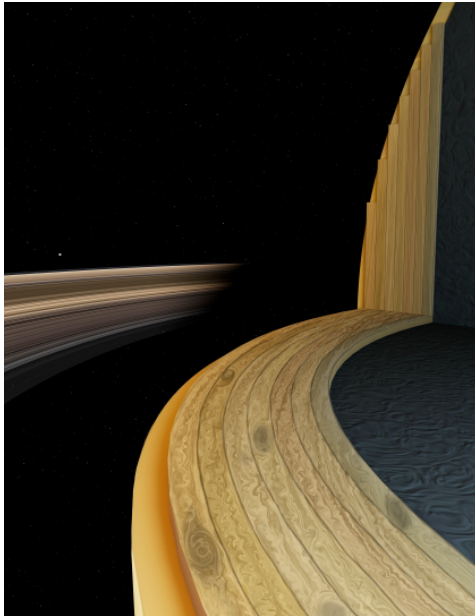


Bending Waves



$$f(\text{Radius, Mass, } \mathbf{Inclination}) = \text{Amplitude}_{\text{BW}}$$

Differential Rotation Region



Results

Type	A	Mass (kg)	Inclination ($^{\circ}$)	Height (km)
DW BW	0.235 0.032	$(6.47 \pm 2.37) \times 10^{15}$	3.19 ± 0.00	3079 ± 2415
DW BW	0.087 0.164	$(3.05 \pm 2.00) \times 10^{15}$	60.06 ± 0.23	47920 ± 32890
DW BW	0.115 0.108	$(4.00 \pm 2.26) \times 10^{15}$	10.77 ± 0.50	10340 ± 6383
DW BW	0.279 0.036	$(2.06 \pm 0.88) \times 10^{16}$	1.25 ± 0.045	1207 ± 826
DW BW	0.436 0.031	$(9.65 \pm 3.25) \times 10^{15}$	2.06 ± 0.047	1985 ± 1400
DW BW	0.126 0.108	$(2.37 \pm 0.98) \times 10^{17}$	8.2 ± 0.00	79.18 ± 36.54

What's Next?

What's Next?

- Differential rotation

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- Differential rotation
- Planetary formation

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- Differential rotation
- Planetary formation
- Additional, smaller anomalies

What's Next?

- Differential rotation
- Planetary formation
- Additional, smaller anomalies
- Other resonances

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- Maryame El Moutamid
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